ENTERPRISE ARCHITECTURE (EA) AND DATABASE PROGRAMMING

LECTURE 5

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CONTENTS:

- Getting to Grips with Architectural Complexity
- Describing Enterprise Architectures
- Pictures, Models, and Semantics

GETTING TO GRIPS WITH ARCHITECTURAL COMPLEXITY

- Companies have long recognized the need for an integrated architectural approach and have developed their own architecture practice. However, they still experience a lack of support in the design, communication, realization, and management of architectures.

GETTING TO GRIPS WITH ARCHITECTURAL COMPLEXITY

- **Design:** When designing architectures, architects should use a common, well-defined vocabulary to avoid misunderstandings and promote clear designs. Such a vocabulary must not just focus on a single architecture domain but should allow for the integration of different types of architectures related to different domains. Next to a common language, architects should be supported in their design activities by providing methodical support, general and organization specific guidelines, best practices, drawing standards, and other means that promote the quality of the architectures. Furthermore, to facilitate the design process, which is iterative and requires changes and updates to architectures, support for tracking architectural decisions and changes is desirable.
- **Communication:** Architectures are shared with various stakeholders within and outside the organization. To facilitate the communication about architectures, it should be possible to visualize precisely the relevant aspects for a particular group of stakeholders. Especially important in this respect is to bring about a successful communication on relations among different domains described by different architectures (e.g., processes vs. applications), since this will often involve multiple groups of stakeholders. Clear communication is also very important in the case of outsourcing of parts of the implementation of an architecture to external organizations.

GETTING TO GRIPS WITH ARCHITECTURAL COMPLEXITY

- **Realization:** To facilitate the realization of architectures and to provide feedback from this realization to the original architectures, links should be established with design activities on a more detailed level, e.g., business process design, information modelling, or software development. Companies use different concepts and tools for these activities, and relations with these should be defined. Furthermore, integration with existing design tools in these domains should be provided.
- **Change:** An architecture often covers a large part of an organization and may be related to several other architectures. Therefore, changes to an architecture may have a profound impact. Assessing the consequences of such changes beforehand and carefully planning the evolution of architectures are therefore very important. Until now, support for this has been virtually non-existent.

GETTING TO GRIPS WITH ARCHITECTURAL COMPLEXITY

- **Compositionality**

In current practice, enterprise architectures often comprise many heterogeneous models and other descriptions, with ill-defined or completely lacking relations, inconsistencies, and a general lack of coherence and vision.

The main driver behind most of the needs identified above is the complexity of architectures, **their relations, and their use**. Many different architectures or architectural views co-exist within an organization. These architectures need to be understood by different stakeholders, each at their own level. The connections and dependencies that exist among these different views make life even more difficult. Management and control of these connected architectures is extremely complex.

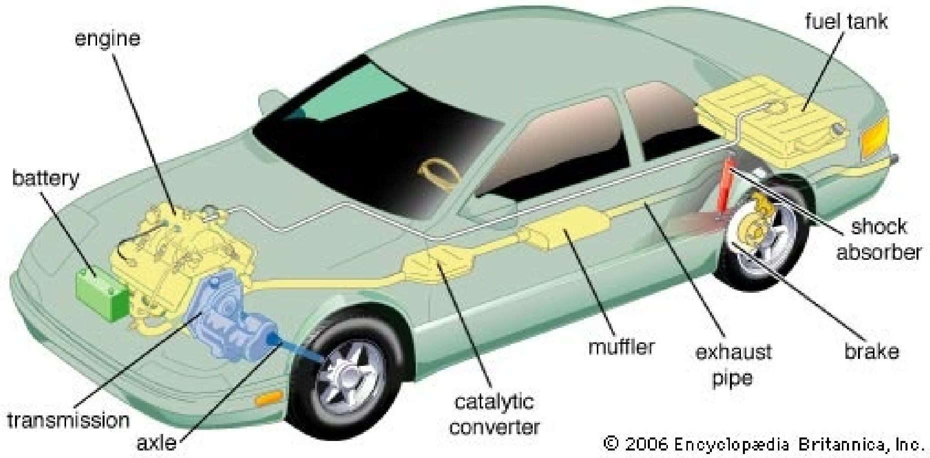
GETTING TO GRIPS WITH ARCHITECTURAL COMPLEXITY

- **Compositionality**

- The standard approach to dealing with the complexity of systems is to use a compositional approach, which distinguishes between parts of a system, and the relations between these parts.

- To understand how a car functions, we first describe the parts of the car such as the engine, the wheels and the air conditioning system, and then we describe the relationship among these parts.

- Likewise, we understand the **information system** of a company as a set of systems and their relations, and we understand a company as a set of **business processes** and their relations.



GETTING TO GRIPS WITH ARCHITECTURAL COMPLEXITY

- **Integration of Architectural Domains**

- **Example 1:** As a first example of an integration problem, consider Fig. 3.1, which contains several architectures. The five architectures may be models expressed in UML, or models from cells of Zachman's architectural framework, or any kind of combination. For instance, there may be a company that has modelled its applications in UML and its business processes in BPMN.
- In all these cases, it is unclear how concepts in one view are related to concepts in another view.
- Moreover, it is unclear whether views are compatible with each other.

GETTING TO GRIPS WITH ARCHITECTURAL COMPLEXITY

- **Integration of Architectural Domains**

This is often called a problem of alignment, and the UML–BPMN example is called a business–IT alignment problem.

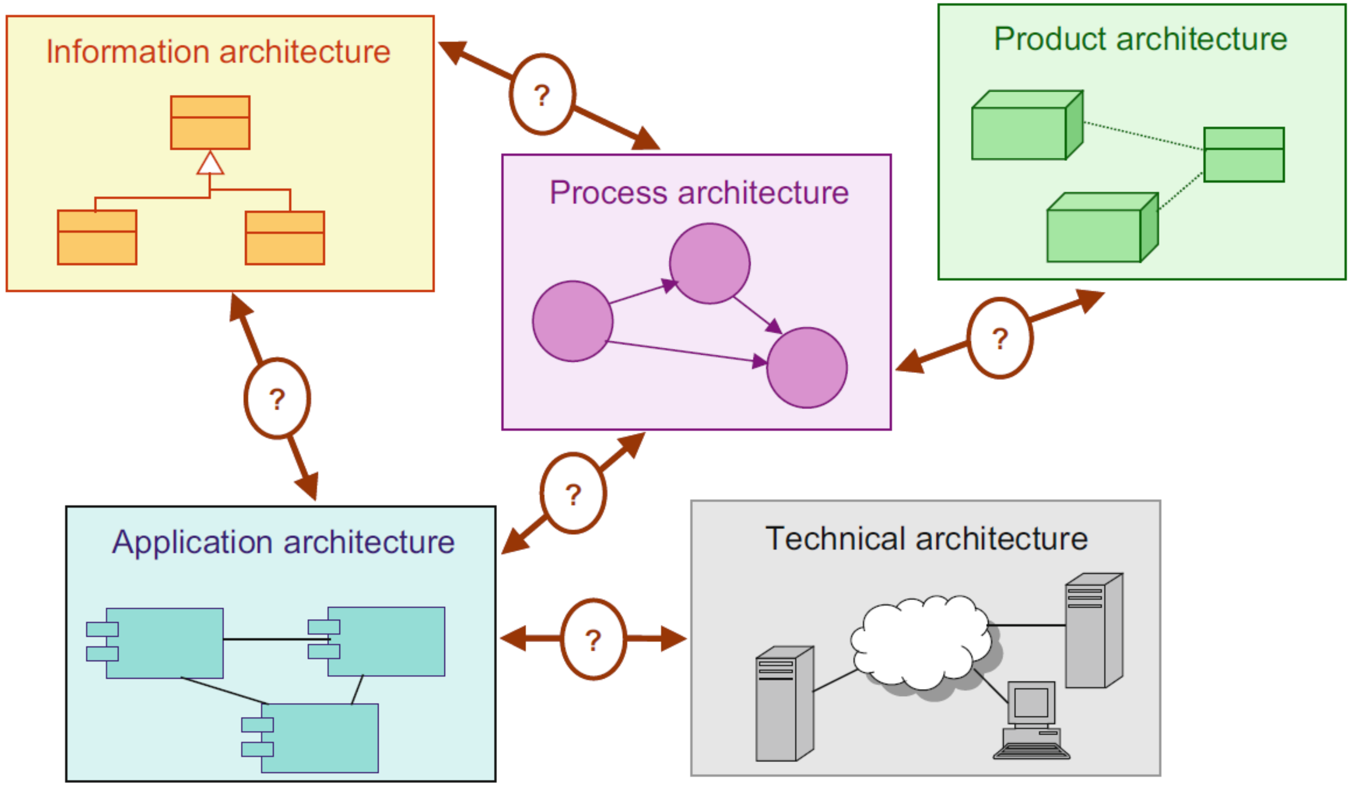


Fig. 3.1 Heterogeneous architectural domains

GETTING TO GRIPS WITH ARCHITECTURAL COMPLEXITY

- **Integration of Architectural Domains**

- **Example 2:** As a second example of an integration problem, consider the first two viewpoints of the IEEE 1471 standard (IEEE Computer Society 2000): the structural viewpoint and the behavioral viewpoint.
- How are structure and behavior related?
- One instance of this problem is how structural concepts like software components are related to behavioral concepts like application functions.

GETTING TO GRIPS WITH ARCHITECTURAL COMPLEXITY

Discussion

According to the below paper describe how **different enterprise architecture domains can be integrated?**

- *Antunes, G., Caetano, A., Bakhshandeh, M., Mayer, R., & Borbinha, J. (2013, June). Using ontologies to integrate multiple enterprise architecture domains. In International Conference on Business Information Systems (pp. 61-72). Springer, Berlin, Heidelberg.*

DESCRIBING ENTERPRISE ARCHITECTURES

- **Describing Enterprise Architectures**

We concentrate on what is essential for enterprise architecture, and therefore we limit ourselves to the core elements of these domains and focus especially on the relations and interactions between them. Precise definitions and constraints will help us to create insight into the complexity of the enterprise architecture and to evade conflicts and inconsistencies between the different domains.

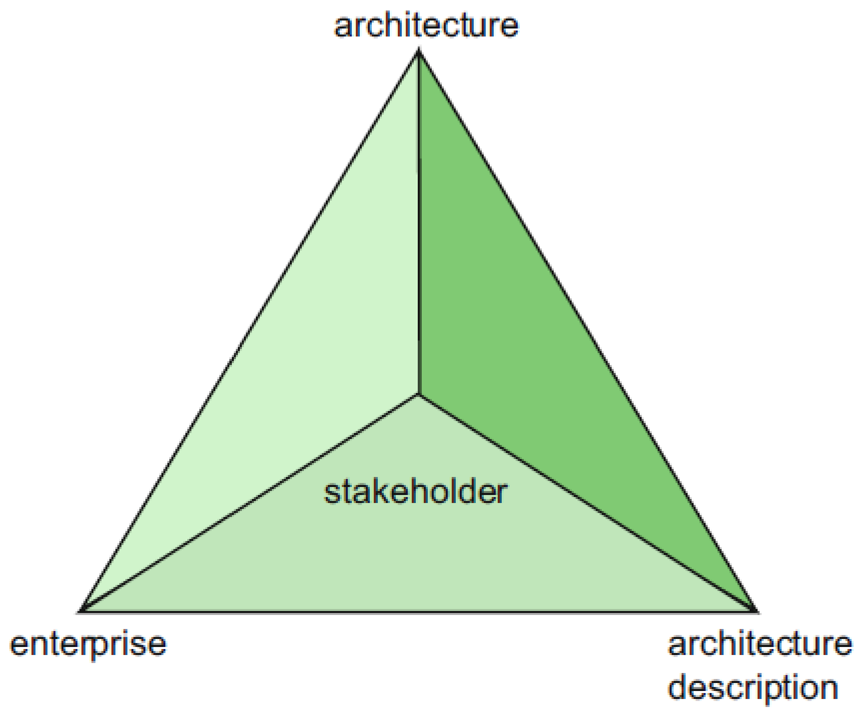
DESCRIBING ENTERPRISE ARCHITECTURES

- **Observing the Universe**

The viewer is a stakeholder of (part of) the organizational, technical, or other systems that make up the enterprise, i.e., the universe that the viewer observes. The conception of this universe then is the architecture of the enterprise.

The representation of this architecture is an architecture description, which may contain models of the architecture, but also, for example, textual descriptions. The underlying relationships between stakeholder, enterprise, architecture and architecture description can be expressed in the form of a tetrahedron, as depicted in Fig. 3.2, which is based on the FRISCO tetrahedron (Falkenberg et al. 1998).

Fig. 3.2 Relationship between enterprise, stakeholder, architecture, and architecture description



DESCRIBING ENTERPRISE ARCHITECTURES

- **Observing Domains**
  - **Domain:** any subset of a conception (being a set of elements) of the universe that is conceived of as being some 'part' or 'aspect' of the universe.
  - **Model:** a purposely abstracted and unambiguous conception of a domain.
  - **Modelling:** the act of purposely abstracting a model from (what is conceived to be) a part of the universe.

DESCRIBING ENTERPRISE ARCHITECTURES

- **Views and Viewpoints**
  - **View:** expresses the architecture of the system of interest from the perspective of one or more stakeholders to address specific concerns, using the conventions established by its viewpoint
  - **Viewpoint:** a specification of the conventions for constructing, interpreting, using and analyzing one type of architecture view

DESCRIBING ENTERPRISE ARCHITECTURES

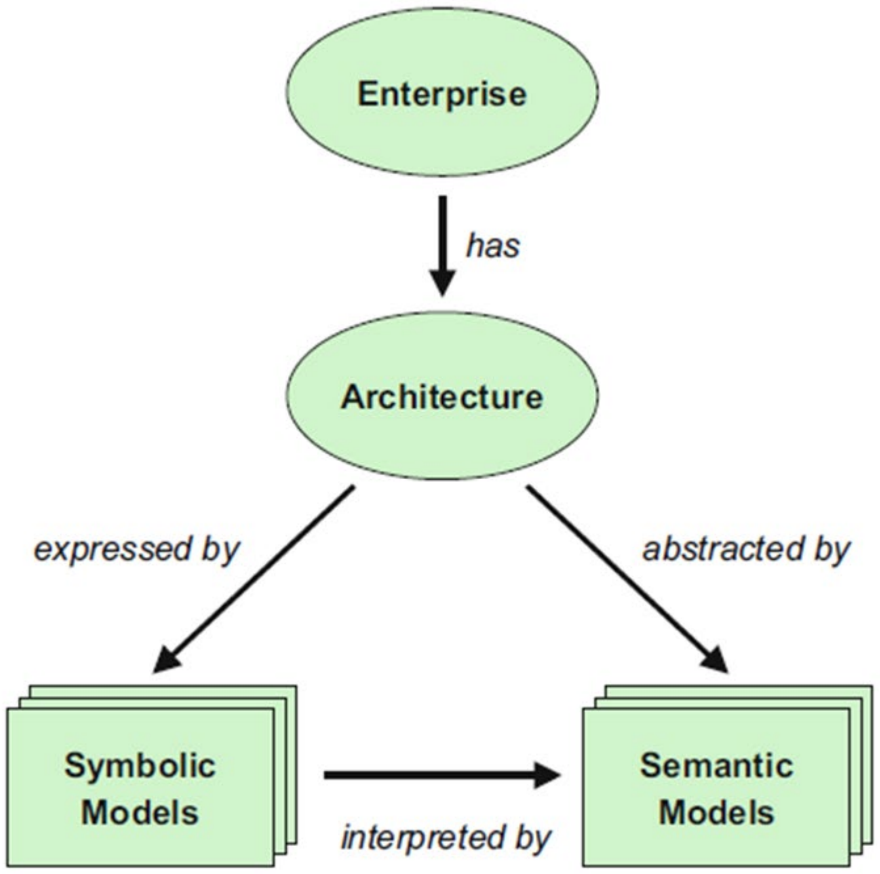
- **Ways of Working**

- **A way of thinking:** articulates the assumptions about the kinds of problem domains, solutions, and modellers involved.
- **A way of modelling:** identifies the core concepts of the language that may be used to denote, analyse, visualise, and/or animate architecture descriptions.
- **A way of communicating:** describes how the abstract concepts from the way of modelling are communicated to human beings, e.g., in terms of a textual or a graphical notation (syntax, style, medium).
- **A way of working:** structures (parts of) the way in which a system is developed. It defines the possible tasks, including subtasks, and ordering of tasks, to be performed as part of the development process. It furthermore provides guidelines and suggestions (heuristics) on how these tasks should be performed.
- **A way of supporting:** the support that is offered by (possibly automated) tools for the handling (creating, presenting, modifying, etc.) of models and views. In general, a way of supporting is supplied in the form of some computerised tool.
- **A way of using:** identifies heuristics that:
  - define situations, classes of stakeholders, and concerns for which a particular model or viewpoint is most suitable;
  - provide guidance in tuning the viewpoint to specific situations, classes of stakeholders, and their concerns.

PICTURES, MODELS, AND SEMANTICS

- **Symbolic Models:** A symbolic model expresses properties of architectures of systems by means of symbols that refer to reality.
- **Semantic Models:** A semantic model is an interpretation of a symbolic model, expressing the meaning of the symbols in that model.

Fig. 3.3 The enterprise, its architecture, symbolic and semantic models



PICTURES, MODELS, AND SEMANTICS

- **Symbolic Models:**

Fig. 3.4 exhibits a structural description of the employees of a company. We need to recall that the above is a syntactic structure; that is, a description of a symbolic model with a signature whose sorts are Employee and Director, and with respective entities related by a relation named Responsible_for.

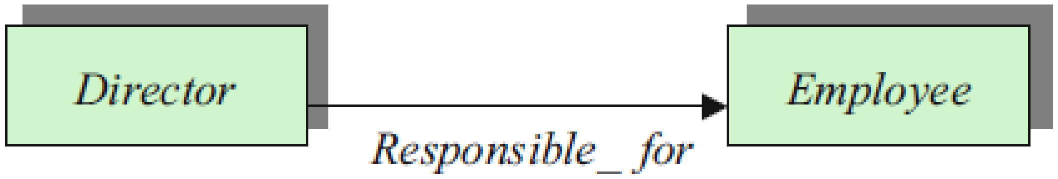


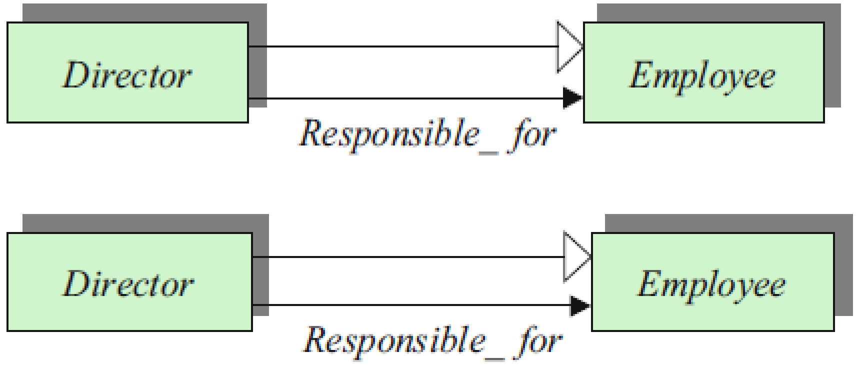
Fig. 3.4 Symbolic model of the director–employee relationship

PICTURES, MODELS, AND SEMANTICS

- **Symbolic Models:**

For example, Fig. 3.5 extends the previous signature with an ‘is-a’ relationship between the sorts Director and Employee (denoted by a UML inheritance relation), intuitively suggesting that every director is also an employee. Moreover, the symbolic model may also contain a set of actions, and the signature a set of action symbols.

Fig. 3.5 Extended symbolic model



PICTURES, MODELS, AND SEMANTICS

- **Semantic Models:** for a symbolic model, we will denote by x:T a variable x which ranges over individuals of sort T. For example, we could use the logical sentence
- ***∃ x : Director. 8y : Employee. Responsible_ for(x, y)***
- to constraint the interpretation of the sort Director to be a non-empty set. Note that since Director is_a Employee, also the interpretation of the latter sort will be non-empty.

PICTURES, MODELS, AND SEMANTICS

Discussion

According to the below paper describe **how symbolic models can be integrated using an architectural language, how integrated models can be updated using the distinction between symbolic models and their visualization, and how semantic models can be integrated?**

- *Arbab, F., de Boer, F., Bonsangue, M., Lankhorst, M., Proper, E., & van der Torre, L. (2007). Integrating architectural models-symbolic, semantic and subjective models in enterprise architecture. Enterprise Modelling and Information Systems Architectures (EMISAJ), 2(1), 40-57.*

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- Antunes, G., Caetano, A., Bakhshandeh, M., Mayer, R., & Borbinha, J. (2013, June). Using ontologies to integrate multiple enterprise architecture domains. In International Conference on Business Information Systems (pp. 61-72). Springer, Berlin, Heidelberg.

- Antunes, G., Caetano, A., Bakhshandeh, M., Mayer, R., & Borbinha, J. (2013, June). Using ontologies to integrate multiple enterprise architecture domains. In International Conference on Business Information Systems (pp. 61-72). Springer, Berlin, Heidelberg.